

Stats 2 : Probability

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1. Experiment & Outcomes

Experiment : An experiment is a process or phenomenon that we wish to study statistically.

- Tossing a coin
- Rolling a die

Outcome : Result of an experiment

- Tossing a coin - Head, Tail
 - Rolling a die - 1, 2, 3, 4, 5 & 6
-

2. Mathematical Theory

In a mathematical theory, we

- precisely define the objects of interest
- make a few assumptions, which are called as axioms
- develop all further results through logical deduction.

Probability Theory : It is a mathematical theory for describing the **likelihood** of events in an experiment. Here, the objects of interest are :

- Sample Space : S
- Event Space : $\mathcal{F}$
- Probability Function $P : \mathcal{F} \rightarrow [0, 1]$

These are defined in the upcoming sections.

3. Sample Space

A sample space is a set that contains all the outcomes.

- Tossing a coin - $\{\text{Heads, Tails}\}$
 - Rolling a die - $\{1, 2, 3, 4, 5, 6\}$
-

4. Events

An event is a collection of outcomes. An event is a subset of sample space.

- Rolling a die - Outcome is less than 4
– $\{1, 2, 3\}$
- Set operations such as union, intersection, and complement can be applied to events

Event Space : The set of all the events, denoted by $\mathcal{F}$, is the power set of the sample space.

5. Probability Function

$P : \mathcal{F} \rightarrow [0, 1]$ is a probability function if it satisfies the following axioms.

- **Axiom-1** : For any event A , $P(A) \geq 0$
- **Axiom-2** : $P(S) = 1$

- **Axiom-3** : If $E_1, \dots, E_n$ are disjoint event, then $P(E_1 \cup \dots \cup E_n) = P(E_1) + \dots + P(E_n)$
-

6. Probability Space

The following tuple is referred to as a probability space.

$$\langle S, \mathcal{F}, P \rangle$$

where,

- S is the sample space of an experiment
 - $\mathcal{F}$ is the power set of the sample
 - P is the probability function
-

7. Properties

Property 1 : Let A be an event. Then, $P(A^C) = 1 - P(A)$

Proof : A and A^C are disjoint. Also, $S = A \cup A^C$.

$$P(S) = 1$$

$$P(A \cup A^C) = 1$$

$$P(A) + P(A^C) = 1$$

$$P(A^C) = 1 - P(A)$$

Property 2 : $P(\phi) = 0$, where ϕ is the null set.

Proof :

$$P(S^C) = 1 - P(S)$$

$$P(\phi) = 1 - P(S)$$

$$P(\phi) = 0$$

Property 3 : If event B is a subset of event A , then, $P(B) \leq P(A)$

Proof : Since $B \subseteq A$,

$$A = B \cup B^C$$

Also, $P(B^C) \geq 0$

$$P(B \cup B^C) = P(A)$$

$$P(B) + P(B^C) = P(A)$$

$$P(B) \leq P(A)$$

Property 4 : If A and B are two events, then

- $P(A) = P(A \cap B) + P(A \setminus B)$
- $P(B) = P(A \cap B) + P(B \setminus A)$

Proof : Set A and set B can be express as a union of two disjoint sets.

$$\begin{aligned} A &= (A \cap B) \cup (A \setminus B) \\ B &= (A \cap B) \cup (B \setminus A) \end{aligned}$$

By **axiom-3**, the given property is true.

Property 5 : If A and B are two events, then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Proof : $A \cup B$ can be expressed as a union of three disjoint set.

$$A \cup B = (A \setminus B) \cup (B \setminus A) \cup (A \cap B)$$

By property-4,

- $P(A) - P(A \cap B) = P(A \setminus B)$
- $P(B) - P(A \cap B) = P(B \setminus A)$

$$\begin{aligned} P(A \cup B) &= P(A \setminus B) + P(B \setminus A) + P(A \cap B) \\ &= P(A) - P(A \cap B) + P(B) - P(A \cap B) + P(A \cap B) \\ &= P(A) + P(B) - P(A \cap B) \end{aligned}$$

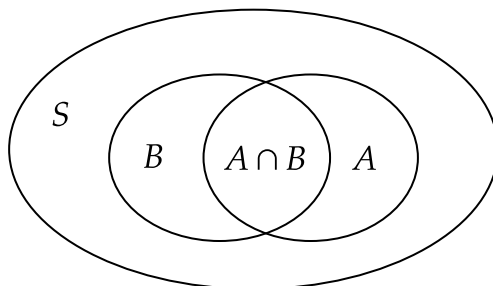
Similarly, one can show that

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

8. Conditional Probability

Given that event B has occurred, the probability of event A occurring is given by

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



9. Mutually Disjoint event

$B_1, \dots, B_n$ are mutually disjoint events if

$$\begin{aligned} B_i \cap B_j &= \phi \\ \forall i, j \in \{1, \dots, n\} \\ &\& \\ i &\neq j \end{aligned}$$

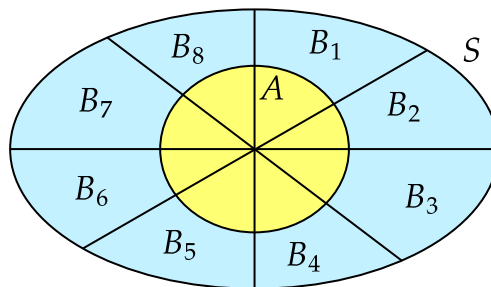
10. Total Probability

Suppose.

$$A = (A \cap B_1) \cup (A \cap B_2) \cup \dots \cup (A \cap B_n)$$

where, $B_1, \dots, B_n$ are mutually disjoint events and if $B_1 \cup \dots \cup B_n = S$ then,

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + \dots + P(A \cap B_n)$$



11. Bayes' Theorem

Let A and B be events with $P(A) > 0$ and $P(B) > 0$. Then,

$$P(A|B) \times P(B) = P(B|A) \times P(A)$$

This is coming from conditional probability

$$P(A|B) \times P(B) = P(A \cap B) = P(B|A) \times P(A)$$

12. Independence of Events

Two events A and B are independent if

$$P(A \cap B) = P(A) \times P(B)$$

Independence of two events implies that the occurrence of one event does not change the chance of occurrence of the other event.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \times P(B)}{P(B)} = P(A)$$

Similarly, $P(B|A) = P(B)$

Note : Independent events need not be disjoint events. If two event M and N are disjoint, then

$$P(M \cap N) = P(\phi) = 0$$

13. Random Variable

A random variable is a function with domain as the sample space and codomain as the real numbers.

Example

- Experiment : Tossing two coins.
- Sample Space : $S = \{HH, HT, TH, TT\}$
- Random Variable : $X : S \rightarrow \mathbb{R}$, denotes the number of heads
- $X(HH) = 2$
- $X(HT) = 1$
- $X(TH) = 1$
- $X(TT) = 0$

14. Countable & Uncountable Sets (Informal)

Defining countable and uncountable sets are beyond the scope of our discussions. We shall give examples of such sets.

- A set is countable if it is either **countably finite** or **countably infinite**
- A set is uncountable if it is **not countable**

Counting : In simple words, counting means arranging elements so each one appears at a definite position (1st, 2nd, 3rd, ...).

Examples :

- The set $\{1, 2, 3, 4\}$ is an example of countably finite set. We can list all the elements and the

number of elements in this set is finite

- The set of all the natural numbers is a countably infinite set. We can count the elements of the set, but there are infinitely many elements
 - The set of all the real numbers, denoted by $\mathbb{R}$, is an uncountable set. Between any two distinct real number, we can find another real number. So, it is not possible to list all the elements of $\mathbb{R}$
 - Any subset of a countable set is a countable set
 - Any interval of the form (a, b) , $[a, b)$, $(a, b]$ or $[a, b]$, where $a < b$ is an example of an uncountable set
-

15. Discrete Random Variable

A discrete random variable takes values in a countable set.

Probability Mass Function (PMF)

- Let $X : S \rightarrow \mathbb{R}$ be a discrete random variable with T as the range set.
- For each $u \in T$, let us define the event A_u as follows

$$A_u = \{v \in S | X(v) = u\}$$

- The probability mass function of X is $f_X : T \rightarrow [0, 1]$, where

$$f_X(u) = P(A_u)$$

Cumulative Distribution Function (CDF) - Discrete

- Let $X : S \rightarrow \mathbb{R}$ be a discrete random variable with T as the range set.
- Let $f_X : T \rightarrow [0, 1]$ be the PMF of X
- For each $u \in T$, let us define a set L_u as follows

$$L_u = \{v \in T | v \leq u\}$$

- The CDF of X is $F_X : \mathbb{R} \rightarrow [0, 1]$, where

$$F_X(u) = \sum_{v \in L_u} f_X(v) = P(X \leq u)$$

- If, for some $u \in \mathbb{R}$, $L_u = \{\}$, then $F_X(u) = 0$
- Suppose, $T_X = \{x_1, \dots, x_n\}$ and $x_1 < x_2 < \dots < x_n$, Then,

$$F_X(x_i) = P(X \leq x_i) = \sum_{k=1}^i f_X(x_k)$$

Support

Let X be a discrete random variable with f_X as its PMF. The support of X is given by the following set.

$$\{x | f_X(x) > 0\}$$

Common Distributions

Distribution	Parameters	Support	PMF
$X \sim \text{Bernoulli}$	$p \in [0, 1]$	$x \in \{0, 1\}$	$f_X(1) = p$ $f_X(0) = (1 - p)$
$X \sim \text{Binomial}$	$n \in \{0, 1, 2, \dots\}$ $p \in [0, 1]$	$x \in \{0, 1, \dots, n\}$	$f_X(x) = {}^nC_x(p)^x(1 - p)^{n-x}$
$X \sim \text{Poisson}$	$\lambda > 0$	$x \in [0, \infty)$	$f_X(x) = \frac{e^{-\lambda} \lambda^x}{x!}$
$X \sim \text{Geometric}$	$p \in [0, 1]$	$x \in \{1, 2, \dots\}$	$f_X(x) = (p)(1 - p)^{x-1}$

Note - A Bernoulli random variable is also called an indicator random variable.

16. Continuous Random Variable

A continuous random variable takes values over an interval and is described by a density function.

Probability Density Function (PDF)

- Let $X : S \rightarrow \mathbb{R}$ be a continuous random variable
- The probability density function of X is $f_X : \mathbb{R} \rightarrow [0, \infty)$
- For any $a, b \in \mathbb{R}$, where $a \leq b$,

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx$$

- Note:

$$P(X = a) = \int_a^a f_X(x) dx = 0$$

Cumulative Distribution Function (CDF) - Continuous

- Let $X : S \rightarrow \mathbb{R}$ be a continuous random variable

- The CDF of X is $F_X : \mathbb{R} \rightarrow [0, 1]$, where

$$F_X(u) = \int_{-\infty}^u f_X(x) dx$$

- $P(a \leq X \leq b)$ is given as follows

$$P(a \leq X \leq b) = F_X(b) - F_X(a) = \int_a^b f_X(x) dx$$

- Relationship between CDF and PDF

$$f_X(x) = \frac{d}{dx} F_X(x)$$

- F_X is a non-decreasing function, taking non-negative values
- As $x \rightarrow -\infty$, $F_X(x) \rightarrow 0$
- As $x \rightarrow 1$, $F_X(x) \rightarrow 1$

Support

Let X be a continuous random variable with f_X as its PDF. The support of X is given by the following set.

$$\{x | f_X(x) > 0\}$$

Common Distributions

Distribution	Parameters	Support	PDF (for support)
$X \sim \text{Uniform}(a, b)$	$a, b \in \mathbb{R}, a \leq b$	$x \in [a, b]$	$f_X(x) = \frac{1}{b-a}$
$X \sim \text{Exponential}(\lambda)$	$\lambda > 0$	$x \in [0, \infty)$	$f_X(x) = \lambda e^{-\lambda x}$
$X \sim \text{Normal}(\mu, \sigma^2)$	$\mu \in \mathbb{R}, \sigma^2 \geq 0$	$x \in \mathbb{R}$	$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

17. Joint Distribution

Discrete Case

X and Y are two discrete random variables with T_X and T_Y as their corresponding range set. The joint PMF of X and Y , denoted by $f_{XY} : T_X \times T_Y \rightarrow [0, 1]$ is given as follows

$$f_{XY}(x, y) = P(X = x, Y = y)$$

Continuous Case

X and Y are two continuous random variables. The joint PDF of X and Y is given by $f_{XY} : \mathbb{R}^2 \rightarrow [0, \infty)$ if

$$P((X, Y) \in A) = \iint_A f_{XY}(x, y) dx dy$$

for every region A in $\mathbb{R}^2$

18. Marginal Distribution

Discrete Case

X and Y are two discrete random variables with f_{XY} as their joint PMF. The range of X is T_X and the range of Y is T_Y . The individual PMF of X and Y are called the marginal PMFs. They are given as follows.

$$f_X(x) = P(X = x) = \sum_{y' \in T_Y} f_{XY}(x, y')$$

$$f_Y(y) = P(Y = y) = \sum_{x' \in T_X} f_{XY}(x', y)$$

Continuous Case

X and Y are two continuous random variables with f_{XY} as their joint PDF. The individual PDF of X and Y are called the marginal PDFs. They are given as follows.

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y') dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x', y) dx$$

19. Conditional Distribution

Discrete Case

X and Y are two discrete random variables with f_X and f_Y as their PMFs. The joint PMF of X and Y is f_{XY} . The conditional PMF of X given that $Y = y'$ is $f_{X|Y=y'}$, where

$$f_{X|Y=y'}(x) = \frac{f_{XY}(x, y')}{f_Y(y')} = P(X = x|Y = y')$$

Continuous Case

X and Y are two continuous random variables with f_X and f_Y as their PDFs. The joint PDF of X and Y is f_{XY} . The conditional PDF of X given that $Y = y'$ is $f_{X|Y=y'}$, where

$$f_{X|Y=y'}(x) = \frac{f_{XY}(x, y')}{f_Y(y')}$$

20. Independent Random Variables

Discrete Case

X and Y are two discrete random variables with f_X and f_Y as their PMFs and T_X and T_Y as their ranges. The joint PMF of X and Y is f_{XY} . X and Y are said to be independent, if

$$f_{XY}(x, y) = f_X(x) \cdot f_Y(y)$$

$$\forall x \in T_X \ \& \ \forall y \in T_Y$$

Continuous Case

X and Y are two continuous random variables with f_X and f_Y as their PDFs. The joint PDF of X and Y is f_{XY} . X and Y are said to be independent, if

$$f_{XY}(x, y) = f_X(x) \cdot f_Y(y)$$

$$\forall (x, y) \in \mathbb{R}^2$$

21. Independent & Identically Distributed (iid) Random Variables

Two random variables X and Y are said to be independent and identically distributed if

- they are independent
- their PMFs are identical (for discrete)
- their PDFs are identical (for continuous)

Example : Suppose we toss two unbiased coins. Let X and Y be the random variables denoting the

results of coin 1 and coin 2 respectively. Let

- $X(\text{Heads}) = 1 = Y(\text{Heads})$
- $X(\text{Tails}) = 1 = Y(\text{Tails})$

Independence

- The result of a coin does not affect the result of the other. Therefore,
 - $f_{X|Y=1}(x) = f_X(x)$
 - $f_{X|Y=0}(x) = f_X(x)$
 - $f_{Y|X=1}(y) = f_Y(y)$
 - $f_{Y|X=0}(y) = f_Y(y)$

Identically Distributed

- The probability of heads is the same for both the coins
 - $f_X(1) = f_Y(1) = 0.5$
- The probability of tails is the same for both the coins
 - $f_X(0) = f_Y(0) = 0.5$

Therefore, X and Y are independent and identically distributed.

22. Expectation & Variance

Discrete Case

X is a discrete random variable with range T_X and PMF f_X .

- Expectation of X

$$E[X] = \sum_{x \in T_X} f_X(x) \cdot x$$

- Variance of X

$$\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2 = \sum_{x \in T_X} [f_X(x) \cdot (x - E[X])^2]$$

Continuous Case

X is a continuous random variable with PDF f_X .

- Expectation of X

$$E[X] = \int_{-\infty}^{\infty} f_X(x) \cdot x \, dx$$

- Variance of X

$$\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2 = \int_{-\infty}^{\infty} f_X(x) \cdot (x - E[X])^2 dx$$

Standard Deviation

The standard deviation of X is given as follows :

$$SD(X) = \sqrt{\text{Var}(X)}$$

Linearity Of Expectation

X and Y are two random variables. Consider $Z = \alpha X + \beta Y$, where $\alpha, \beta \in \mathbb{R}$

$$\begin{aligned} E[Z] &= E[\alpha X + \beta Y] \\ &= \alpha E[X] + \beta E[Y] \end{aligned}$$

Independence & Expectation

Suppose, X and Y are two independent random variables. Then

$$E[XY] = E[X] \cdot E[Y]$$

Proof : X & Y are discrete. Range of X is T_X and range of Y is T_Y

$$\begin{aligned} E[XY] &= \sum_{x \in T_X} \sum_{y \in T_Y} f_{XY}(x, y) \cdot xy \\ &= \sum_{x \in T_X} \sum_{y \in T_Y} f_X(x) \cdot f_Y(y) \cdot xy \\ &= \sum_{x \in T_X} \left[f_X(x) \cdot x \left(\sum_{y \in T_Y} f_Y(y) \cdot y \right) \right] \\ &= \sum_{x \in T_X} E[Y] \cdot (f_X(x) \cdot x) \\ &= E[Y] \sum_{x \in T_X} f_X(x) \cdot x \\ &= E[X] \cdot E[Y] \end{aligned}$$

Proof : X & Y are continuous.

$$\begin{aligned} E[XY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) \cdot xy \, dy \, dx \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(y) \cdot xy \, dy \, dx \end{aligned}$$

$$\begin{aligned}
&= \int_{-\infty}^{\infty} f_X(x) \cdot x \left[\int_{-\infty}^{\infty} f_Y(y) \cdot y \, dy \right] dx \\
&= \int_{-\infty}^{\infty} E[Y] \cdot (f_X(x) \cdot x) \, dx \\
&= E[Y] \int_{-\infty}^{\infty} f_X(x) \cdot x \, dx \\
&= E[X] \cdot E[Y]
\end{aligned}$$

If X and Y are independent, then $E[XY] = E[X] \cdot E[Y]$. However, having $E[XY] = E[X] \cdot E[Y]$ does not guarantee independence between X and Y .

Independence and Variance

If two random variables X and Y are independent, then

$$Var(X + Y) = Var(X) + Var(Y)$$

Proof :

$$\begin{aligned}
Var(X + Y) &= E[(X + Y)^2] - (E[X + Y])^2 \\
&= E[X^2 + Y^2 + 2XY] - (E[X] + E[Y])^2 \\
&= E[X^2] + E[Y^2] + 2 \cdot E[XY] - (E[X])^2 - (E[Y])^2 - 2 \cdot E[X] \cdot E[Y] \\
&= [E[X^2] - (E[X])^2] + [E[Y^2] - (E[Y])^2] + 2(E[XY] - E[X] \cdot E[Y]) \\
&= Var(X) + Var(Y) + 2(E[X] \cdot E[Y] - E[X] \cdot E[Y]) \\
&= Var(X) + Var(Y)
\end{aligned}$$

A note on simplification,

- $E[XY] = E[X] \cdot E[Y]$, since X and Y are independent.

Note : If X and Y are independent, then $Var(X + Y) = Var(X) + Var(Y)$, but the converse need not be true.

Conditional Expectation (Discrete)

Suppose, X and Y are two discrete random variables with f_X and f_Y as their corresponding PMF. Let T_X be the range of X . The expectation of X , given $Y = y'$ is

$$E[X|Y = y'] = \sum_{x \in T_X} x \cdot f_{X|Y=y'}(x)$$

Conditional Expectation (Continuous)

Suppose, X and Y are two continuous random variables with f_X and f_Y as their corresponding PDF. The expectation of X , given $Y = y'$ is

$$E[X|Y = y'] = \int_{-\infty}^{\infty} x \cdot f_{X|Y=y'}(x) dx$$

Expectation & Variance of Common Distributions

DISTRIBUTION	EXPECTATION	VARIANCE
Bernoulli(p)	p	$p(1 - p)$
Binomial(n, p)	np	$np(1 - p)$
Poisson(λ)	λ	λ
Geometric(p)	$1 / p$	$(1 - p) / p^2$
Uniform(a, b)	$(a + b) / 2$	$(b - a)^2 / 12$
Exponential(λ)	$1 / \lambda$	$1 / \lambda^2$
Normal(μ, σ^2)	μ	σ^2

23. Covariance & Correlation

Covariance

X and Y are two random variables. Their covariance is given as follows :

$$\begin{aligned} Cov(X, Y) &= E[(X - E[X])(Y - E[Y])] \\ &= E[XY - X \cdot E[Y] - Y \cdot E[X] + E[X] \cdot E[Y]] \\ &= E[XY] - E[Y] \cdot E[X] - E[X] \cdot E[Y] + E[X] \cdot E[Y] \\ &= E[XY] - E[X] \cdot E[Y] \end{aligned}$$

- $Cov(X, Y) = Cov(Y, X)$
- $Cov(X, X) = Var(X)$
- $Cov(X, Y) = 0$ implies that the random variables X and Y are uncorrelated

- $Var(X + Y) = Var(X) + Var(Y) + 2 \cdot Cov(X, Y)$

Linearity Of Covariance

X, Y and Z are three random variables. Let $\alpha, \beta \in \mathbb{R}$. Then

- $Cov(X, \alpha Y + \beta Z) = \alpha Cov(X, Y) + \beta Cov(X, Z)$
- $Cov(\alpha X + \beta Y, Z) = \alpha Cov(X, Z) + \beta Cov(Y, Z)$

Correlation

X and Y are two random variables. Their correlation is given as follows :

$$Corr(X, Y) = \frac{Cov(X, Y)}{\sqrt{Var(X) \cdot Var(Y)}}$$

Independence & Covariance

Suppose, X and Y are two independent random variables. Then

$$Cov(X, Y) = 0$$

Proof :

$$\begin{aligned} Cov(X, Y) &= E[XY] - E[X] \cdot E[Y] \\ &= E[X] \cdot E[Y] - E[X] \cdot E[Y] \\ &= 0 \end{aligned}$$

While independence implies that $Cov(X, Y) = 0$, $Cov(X, Y) = 0$ does not imply that X and Y are independent.
